

Oshin Tiwari

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AI/ML Engineer | GenAI, NLP | Predictive Modelling

3+ years of experience delivering high-performance models in production environments. Specialized in LLM fine-tuning, RAG, and inference acceleration using PyTorch, TensorFlow, and FastAPI. Skilled in building scalable ML pipelines on AWS and GCP, with a focus on model packaging, latency reduction, and robust evaluation. Adept at bridging data preprocessing, model training, and deployment to create maintainable, efficient systems aligned with real-world constraints.

Technical Skills

- **Programming:** Python, SQL, R, Bash
- **Machine Learning & AI:** Supervised & Unsupervised Learning, Deep Learning (CNN, RNN, Transformers), Generative AI, LLMs (GPT, BERT), Recommendation Systems, NLP, Information Retrieval
- **Big Data & Cloud:** GCP (Vertex AI, BigQuery, Cloud Dataflow), AWS (Sagemaker, S3), Snowflake, Spark, Databricks
- **ML Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas
- **MLOps & Model Deployment:** FastAPI, GitLab, CI/CD, Streamlit, Docker, Kubernetes
- **Data Processing & Visualization:** Power BI, Tableau, Matplotlib, Seaborn, Nomic
- **Specialized Tools & Technologies:** Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Crew AI

Experience

AI Engineer

SmarterX | Boulder, CO

Apr 2024 – Apr 2025

Led AI-driven initiatives as a **Data Lead**, applying machine learning solutions to regulatory compliance, product classification, and automation, with a focus on **retail** and **supply chain optimization**.

- Built a **production-grade classification pipeline** using **PyTorch** and **GCP Vertex AI**, achieving 92% accuracy across 10k+ product SKUs and supporting real-time inference at scale.
- Deployed AI-driven automation tools with **FastAPI** and **LangChain**, improving document classification efficiency.
- Designed and deployed **anomaly detection models** to identify rare failure modes and data drift across large, noisy operational datasets, improving early alerting and reliability by 20%.
- Constructed **multimodal LLM pipelines** using **RAG** to extract structured content from unstructured PDF documents, reducing processing time by 40%.
- Deployed **zero-shot classification workflows** to address limited labeled data, integrating embeddings for real-time data enrichment and product categorization, boosting search efficiency by 30%.
- Developed lightweight APIs with **FastAPI** to serve **high-throughput inference models**, enabling faster integration and deployment into production workflows.
- Built and maintained internal tools for **batch evaluation** and **visualization** of model outputs, improving experimentation velocity and model interpretability across teams.

AI Data Intern

SmarterX | Boulder, CO

Jan 2024 – Mar 2024

- Developed and deployed an **NLP-driven document classification** pipeline to automate regulatory compliance checks, boosting classification accuracy by 20%.
- Applied **statistical modeling** (regression, GLM) to evaluate AI performance and optimize classification thresholds, improving precision/recall balance and achieving a 25% accuracy gain.
- Leveraged **SQL vector database** (Snowflake) for scalable AI storage and retrieval.
- Optimized data preprocessing and **feature engineering pipelines**, reducing model training time by 30% and speeding up inference for large-scale product dataset.

Graduate Research Assistant | Research Project: [Comparative Analysis of SegNet vs. U-Net](#)

Colorado State University | Fort Collins, CO

May 2022 – Dec 2023

- Conducted independent research in applying **U-Net and SegNet architectures** within a GAN framework for **biomedical image translation** tasks, emphasizing architectural efficiency and performance analysis.
- Analyzed segmentation task performance using the Oxford-IIIT Pet Dataset, SegNet reduced **training epochs** by 50% while maintaining comparable accuracy with U-Net.
- **Identified trade-offs** in memory efficiency and training time for SegNet versus U-Net's superior visual fidelity in certain translation results, leading to recommendations for task-specific use cases.
- Mentored 200+ undergraduate students, as a GTA, by delivering engaging instruction and lab sessions in electronics, physics, and applied mathematics (**calculus, statistics, probability**).

Data Science Intern

Behavior LLC | Remote

Aug 2022 – Dec 2022

- Designed an **XGBoost-based model** with time-series forecasting to detect potential relapse indicators, reducing crisis intervention delays by 25%.
- Automated data preprocessing and **ETL workflows**, saving 32 hours of manual effort monthly.
- Applied **unsupervised learning techniques** (e.g., autoencoders, clustering) to detect irregularities in large-scale patient health dataset, enabling early intervention and improving outcomes.
- Implemented interactive **Tableau dashboards** for clinicians, visualizing patient risk scores and behavioral patterns, enabling personalized intervention strategies.

Projects

Deep Learning for Road Segmentation & Autonomous Pathfinding (NASA, AWS, USDA Hackathon Winner) ([Link](#))

- Built a U-Net segmentation model in PyTorch using LiDAR-derived DEM and hillshade .tif imagery; trained on SageMaker with sliding window inference to handle large geospatial images.
- Developed a Bidirectional A* algorithm in Python with NetworkX to compute optimal rover paths, reducing traversal time by 75%.
- Integrated model predictions with shapefiles to reconstruct road graphs; enabled real-time navigation for simulated rescue scenarios.

AI Chatbot with Natural Language Processing (NLP) in Python ([Link](#))

- Built an AI-powered chatbot using pre-trained Transformer models (GPT, BERT) and speech recognition APIs for dynamic, real-time conversation.
- Implemented natural language understanding (NLU) and response generation using NLTK, spaCy, and Hugging Face Transformers, enhancing intent detection, text classification, and entity recognition.
- Integrated FastAPI & Streamlit for real-time inference, ensuring a scalable, lightweight deployment framework.

Audio Classification & Keyword Spotting using Machine Learning ([Link](#))

- Built a CNN-based classifier for audio signal categorization, leveraging MFCC feature extraction for improved speech recognition.
- Applied Named Entity Recognition (NER) and Part-of-Speech (POS) tagging to classify spoken content and detect keywords.
- Optimized the model for low-latency inference, achieving 96% classification accuracy for real-world speech datasets.

Education

Master of Science (M.S.), Computer Engineering

Colorado State University, Fort Collins, CO

Aug 2021 – Dec 2023

Bachelor of Technology (B.Tech.), Electronics & Instrumentation Engineering

Shri G. S. Institute of Technology and Science, Indore, India

Aug 2017 – May 2021